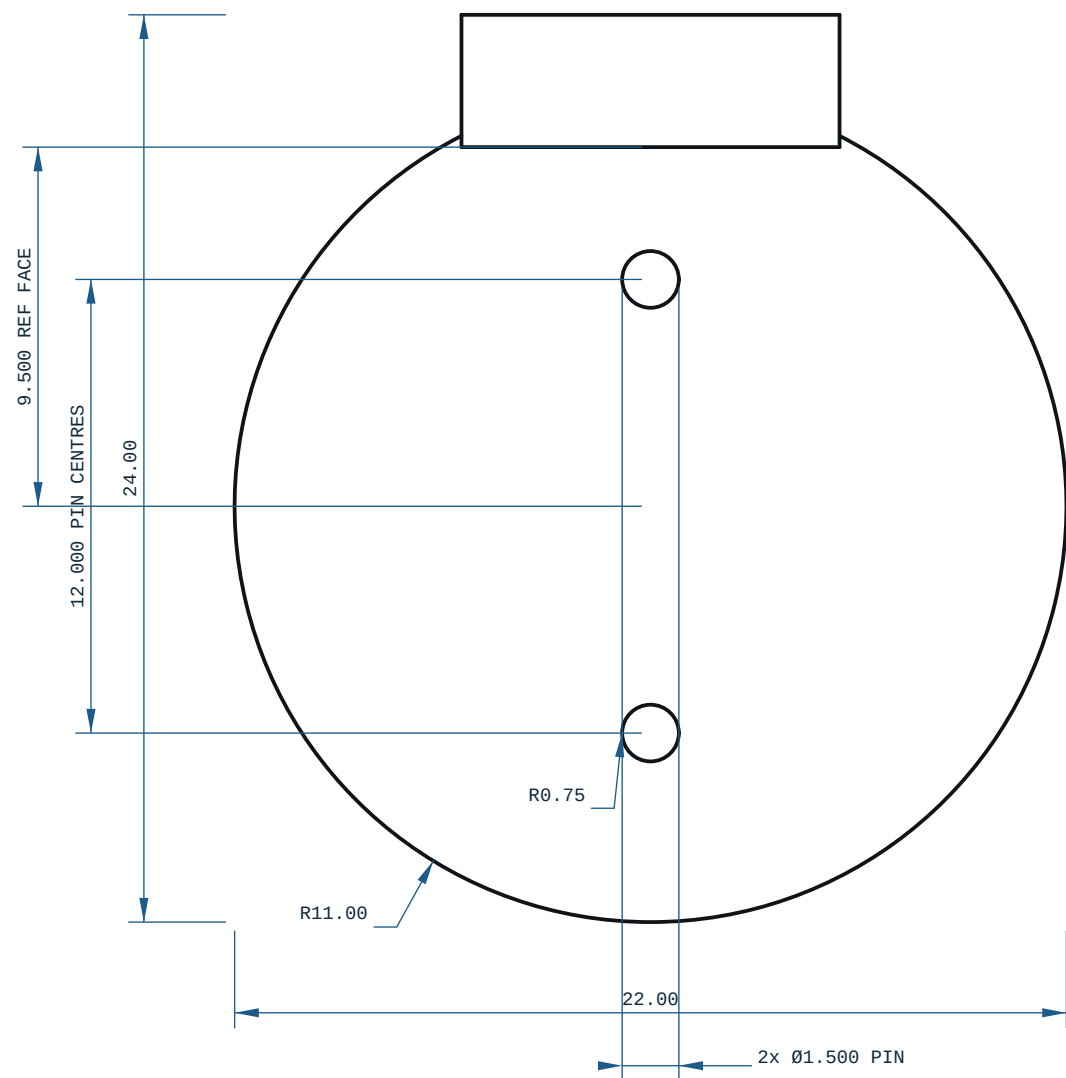
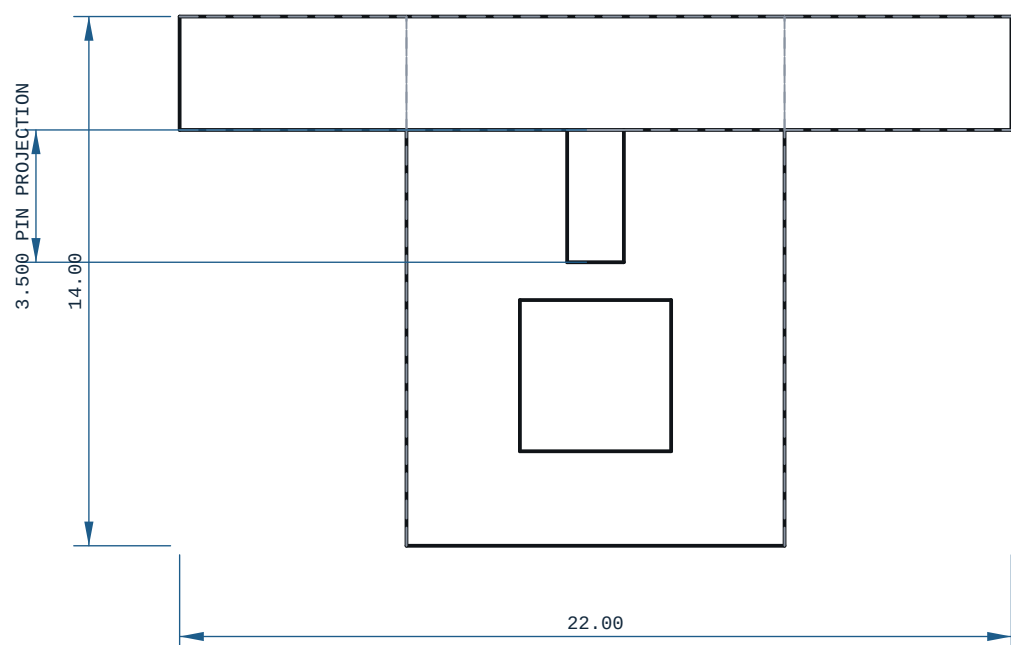




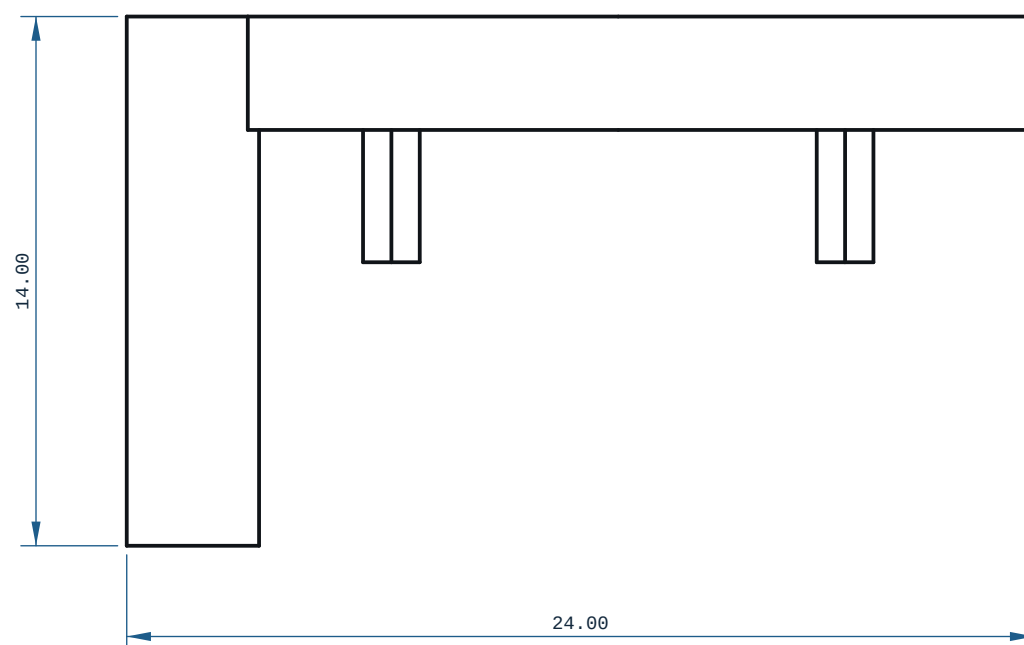
TOP VIEW

X ?      Y ?



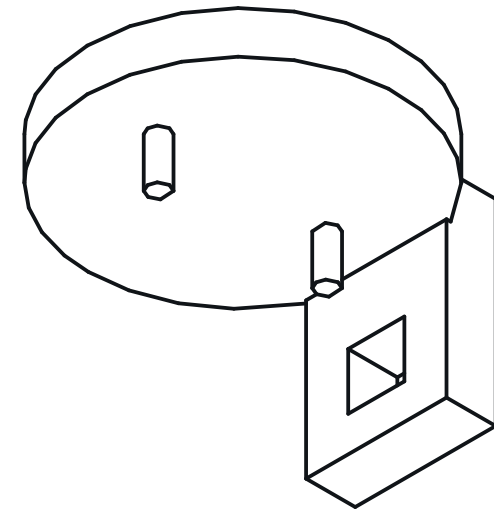
FRONT VIEW

-X ?      Z ?



RIGHT VIEW

-Y ?      Z ?



ISOMETRIC VIEW

22.00 x 24.00 x 14.00 mm

## NOTES

1. ALL DIMENSIONS IN mm AND DEGREES. EVERY DIMENSION AND EVERY HOLE CALLOUT WAS MEASURED OFF THE SOLID.
2. THINNEST WALL, WHOLE SOLID (RAY MINIMUM):  
1.5000 mm, AT X=-0.254 Y=-6.706 Z=-3.208 ?  
MEASURED OVER THE WHOLE SOLID BY RAY+EXACT AT  
0.4810 mm SAMPLE SPACING (A FEATURE NARROWER  
THAN THAT CAN BE MISSED). THIS IS A DISTANCE  
THROUGH MATERIAL, NOT A WALL: AT AN EDGE, A  
FILLET RUN-OUT OR A CHAMFER TIP IT IS REAL AND  
UNPRINTABLE BY DEFINITION. DO NOT JUDGE  
PRINTABILITY BY IT ? THE NEXT NOTE IS THE ONE  
THAT DOES.
3. PRINTABLE MINIMUM WALL: 1.4995 mm, AT X=0.669  
Y=-6.339 Z=-0.875 ? MEASURED AS THE SMALLEST  
MEDIAN WALL OVER A NEIGHBOURHOOD OF 1.4430 mm  
(>= 8 SAMPLES), RAYS CAPPED AT 4.000 mm, BY  
cecad.inspect.printable.wall\_detail. AGAINST  
THE 0.8000 mm FLOOR (2 PERIMETERS @ 0.4 mm  
NOZZLE, cecad/printed.py:801): VERDICT PASS
4. GENERAL TOLERANCE: CANNOT DETERMINE ? THIS PART  
DECLARES NO PROCESS THIS REPO CAN READ.
5. FRONT VIEW: hidden lines KEPT: without them 2  
of 189 edge samples have no ink within 0.35 mm  
(worst 1.400 mm) and the feature would be  
missing from the sheet.
6. TOP VIEW: hidden lines SUPPRESSED: all 183 edge  
samples still land on drawn ink without them.
7. RIGHT VIEW: hidden lines SUPPRESSED: all 183  
edge samples still land on drawn ink without  
them.
8. PINS ARE BOSSES, NOT HOLES, so they carry no  
hole-table row: 2x 01.500 x 3.500 into the  
horn's 1.60 tapped holes = 0.100 mm play =  
0.955 deg of index play. INDEX GAUGE, NOT AN  
ABSOLUTE ZERO.

## PRINT / DFM

1. **ORIENTATION:** laid flat (22 x 24 x 14 mm)
2. **THINNESS WALL, WHOLE SOLID (RAY MINIMUM):**  
1.5000 mm ? a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
3. **PRINTABLE MINIMUM WALL:** 1.4995 mm (smallest median wall over a 1.4430 mm neighbourhood, `cecad.inspect.printable_wall_detail`) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle):  
**VERDICT PASS**
4. **FILAMENT:** PLA, ~2 g (modelled), 70 layers @ 0.2 mm

|                                                |                          |                                                                                       |                           |
|------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------|---------------------------|
| MICRODUCK - JIG - HORN - ZERO                  |                          |  | REV<br>A                  |
| MATERIAL<br>PLA                                |                          | PROCESS<br>-                                                                          | SCALE<br>5:1              |
| CATEGORY<br>-                                  | GENERAL TOL<br>SEE NOTES | FINISH<br>AS PRINTED                                                                  | SHEET<br>A2 1/1           |
| VOLUME (MEASURED)<br>1.55 cm3                  |                          | MASS (ASSUMES PLA)<br>1.9 g                                                           | UNITS<br>mm / deg         |
| SIZE (BBOX, MEASURED)<br>22.00 x 24.00 x 14.00 |                          | DRAWN<br>ce-cad 2026-09-03                                                            | PROJECTION<br>THIRD ANGLE |
| SOURCE (DESIGN FILE)<br>tools/jigs.py          |                          |                                                                                       |                           |